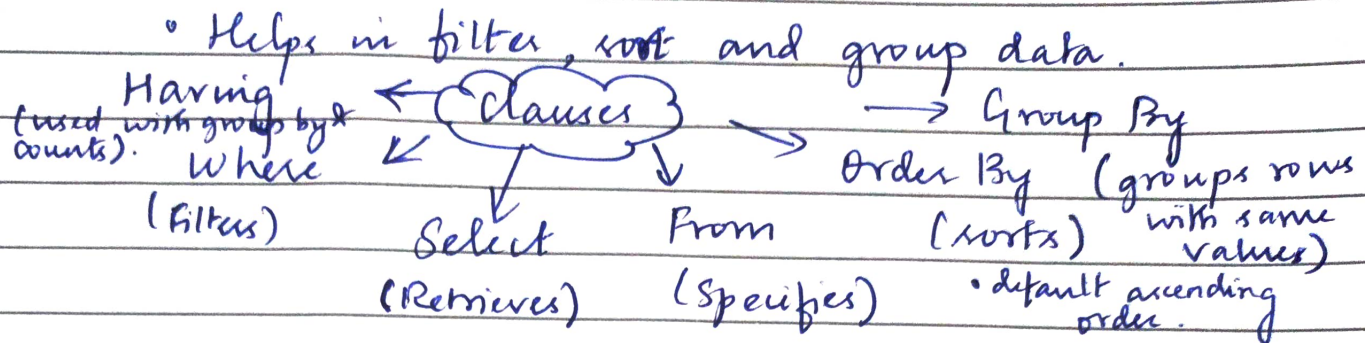


⇒ commit - → to save changes

→ Used when using select statements (mostly).

* Clauses :



* Functions :

- (1) Aggregate functions (sum, count, min, max, avg)
- (2) String functions (upper, lower, length)
- (3) Date functions (current date, now)

* Java Database Connectivity (JDBC)

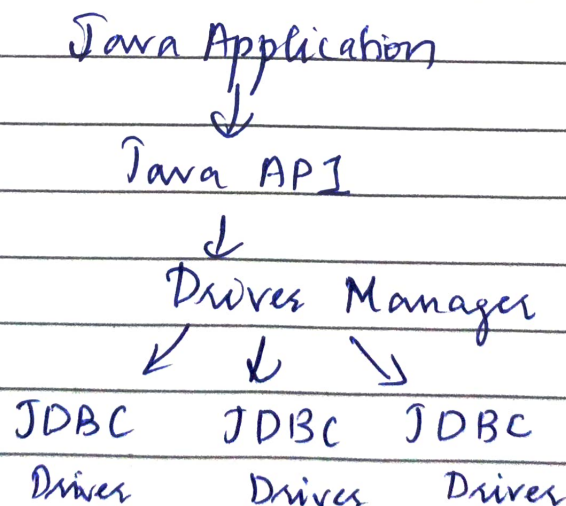


Fig: Flowchart of JDBC Architecture.

1. Create Database

Purpose

Creates a new database to store tables.

Query:

```
CREATE DATABASE school;
```

2. Select Database

Purpose

Chooses the database where all operations will be performed.

Query:

```
USE school;
```

3. Create Table

Purpose

Creates a new table with columns and their data types.

Queries:

Table Teacher:

```
CREATE TABLE teacher (  
    name VARCHAR(10),  
    age INT,  
    department VARCHAR(20)  
);
```

Table student:

```
CREATE TABLE student (  
    name VARCHAR(10),  
    roll_no INT PRIMARY KEY,  
    age INT,  
    department VARCHAR(20)  
);
```

->Concepts Used here are the following:

- **VARCHAR** → Stores text up to n characters.
- **INT** → Stores integer values.
- **PRIMARY KEY** → Uniquely identifies each record and cannot be NULL.

4. ALTER TABLE

Used to modify the structure of an existing table.

A. Add Column

Purpose

Adds a new column to a table.

Query:

ALTER TABLE student

ADD COLUMN degree VARCHAR(100);

B. Rename Column

Purpose

Changes the column name.

Query:

ALTER TABLE student

CHANGE department dept VARCHAR(20);

5. INSERT Statement

Purpose

Adds records into a table.

Queries:

Teacher:

INSERT INTO teacher VALUES ('sri', 24, 'cse');

Student:

INSERT INTO student VALUES ('Nanthu', 22, 24, 'cse', 'B.E');

6. SELECT Statement

Purpose

Displays data stored in a table.

Query:

Teacher:

SELECT * FROM teacher;

Result:

Result Grid				Filter Rows:
	name	age	department	
▶	sri	24	cse	

Student:

SELECT * FROM student;

Result:

Result Grid						Filter Rows:
	name	roll_no	age	dept	degree	
▶	Nanthu	22	24	cse	B.E	